EMCal calibration update: data/sim comparisons

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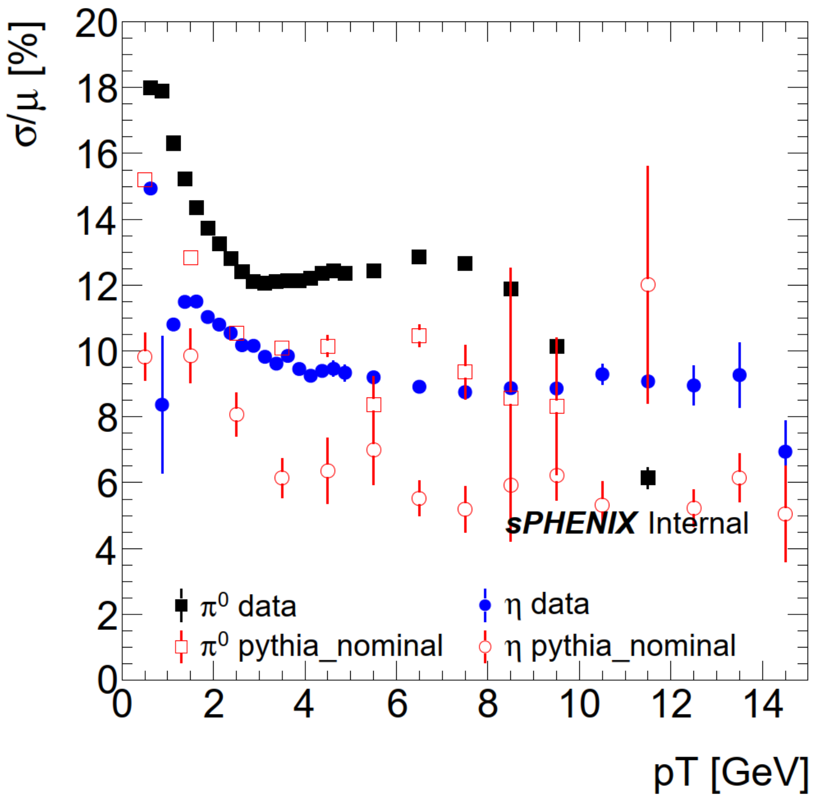
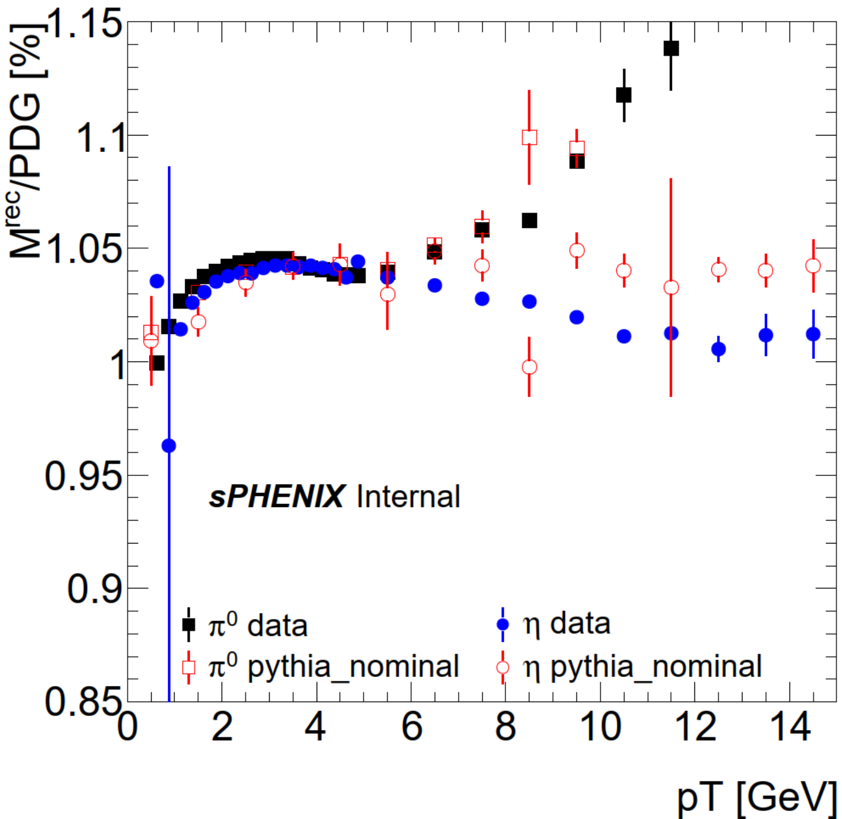
Selections

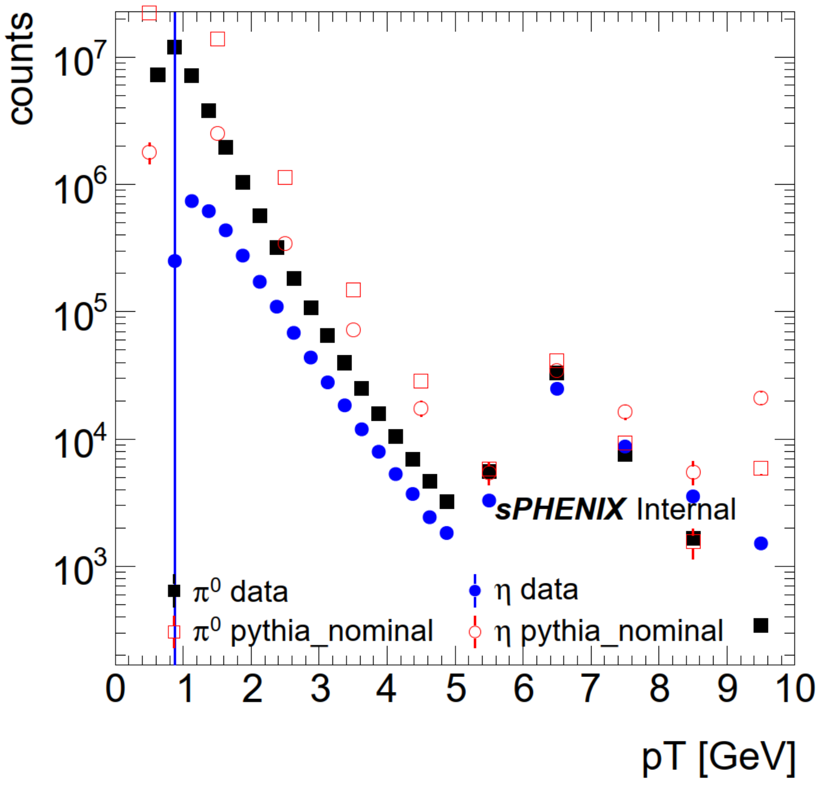
- **Data: ana509, GRL**
- **Pythia run 28: MB, JetXX, Jet12+MB (double interactions)**
- **MBD vertex |z| < 20 cm**
- **Cluster selections**
  - pT1,2 > 0.25 GeV
  - Asym < 0.6

Data pythia comparison

- ➢Pythia: MB + jet5 (pT > 6 GeV) + jet8 (pT > 9 GeV) +jet12 (pT > 12 GeV)
- ➢Data is much wider than sim…
- ➢It is unclear why, and thus we are attempting to smear the MC

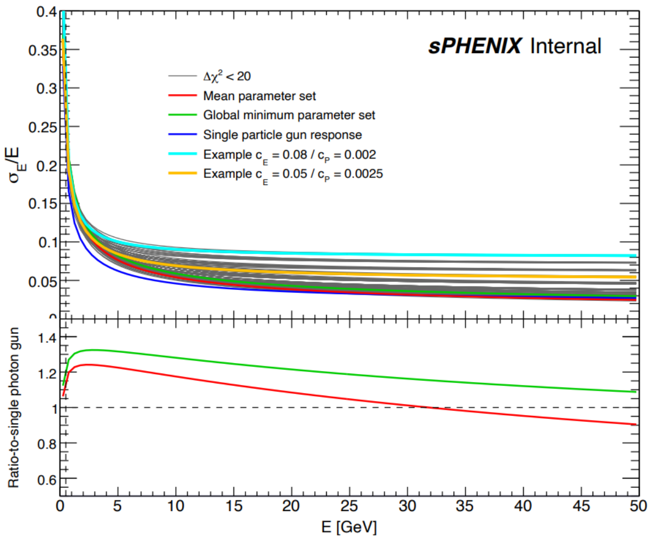


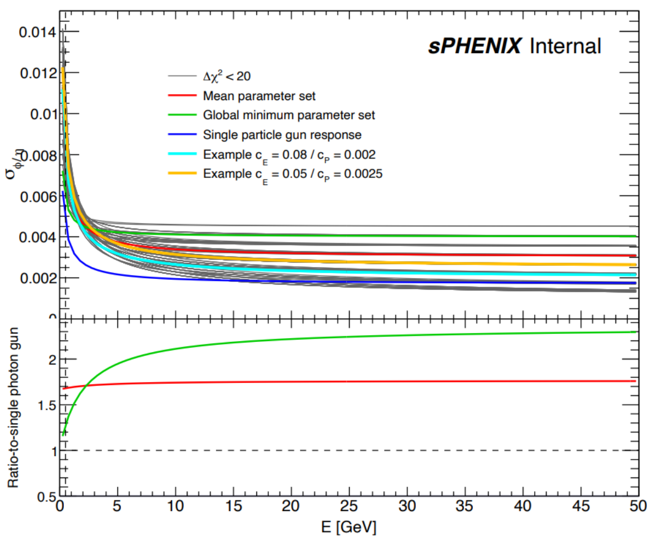


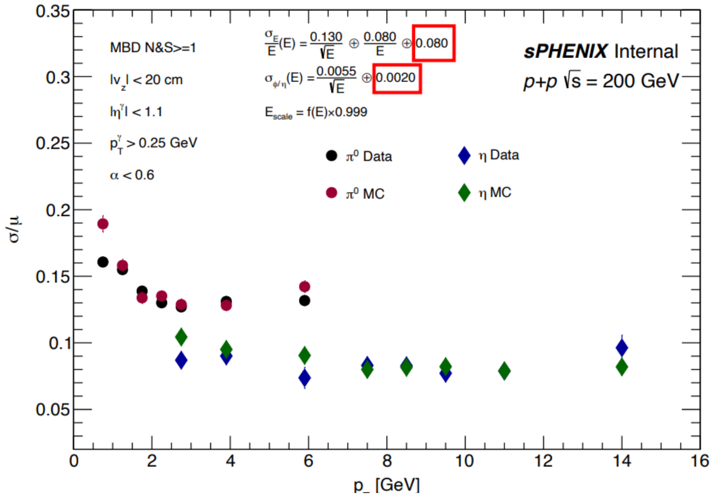


Data-driven extraction of resolution

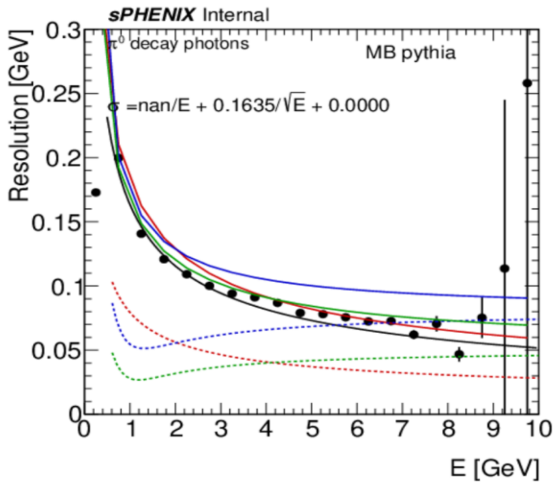
- ➤ https://indico.bnl.gov/event/31859/contributions/120756/attachments/68548/117758/EMCal_toyMC_calib_20260302.pdf

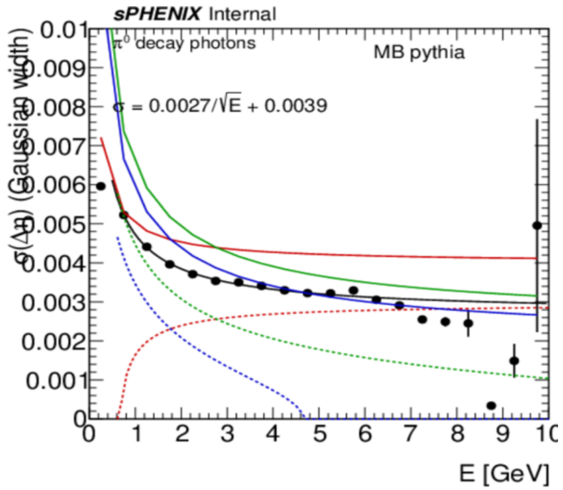






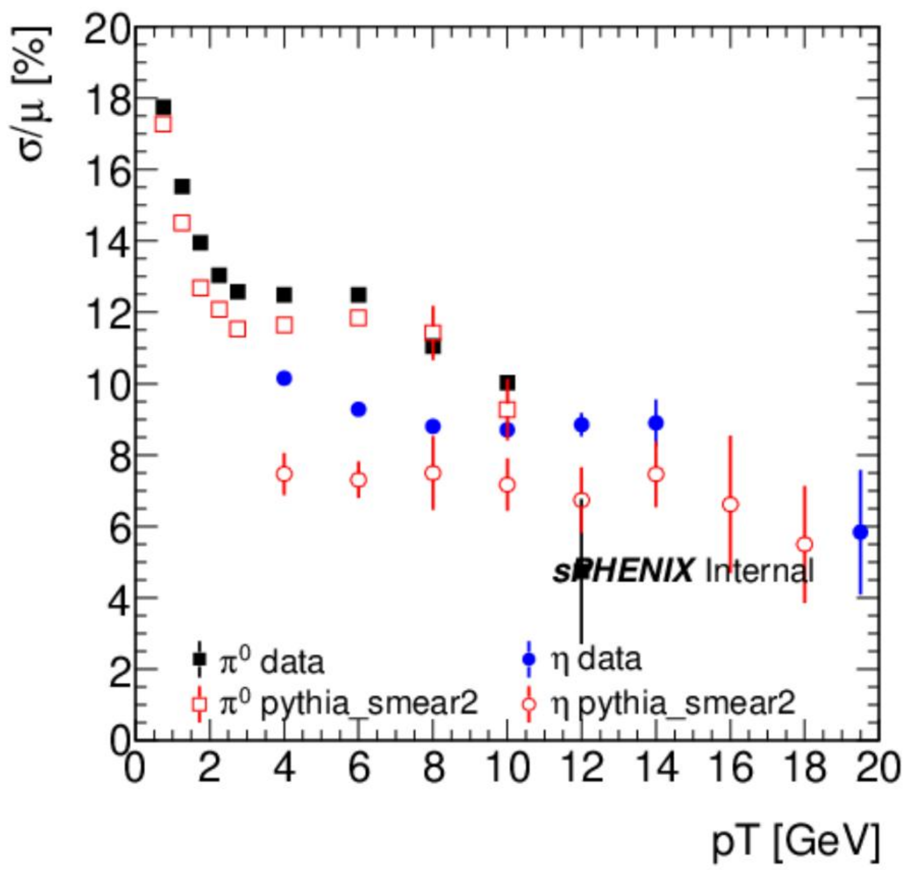
- ➤ **Measure the energy and position resolution in MC to find how much additional smearing is needed to reproduce the data**

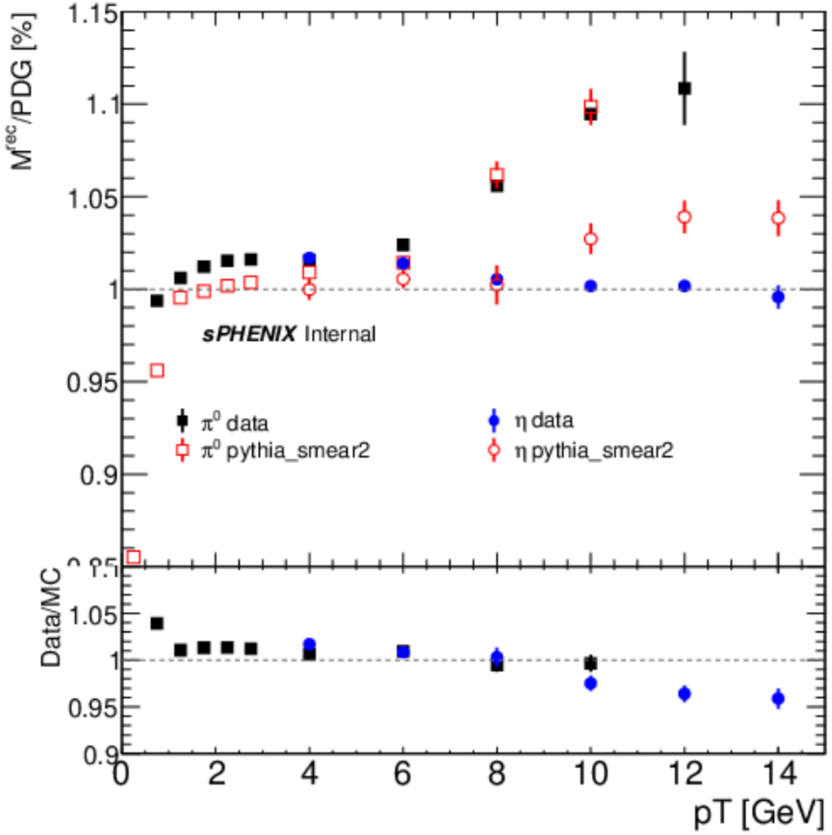




Post smearing (first attempt) data/MC

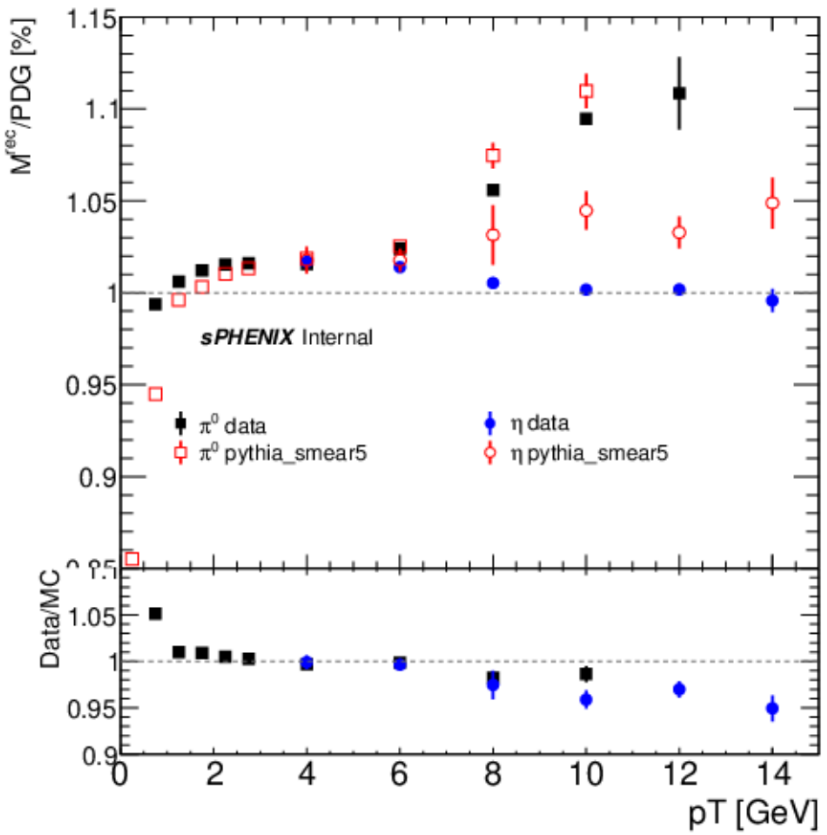
- ➢This well defined process did not seem to work and the smearing factors was insufficient to reproduce the data

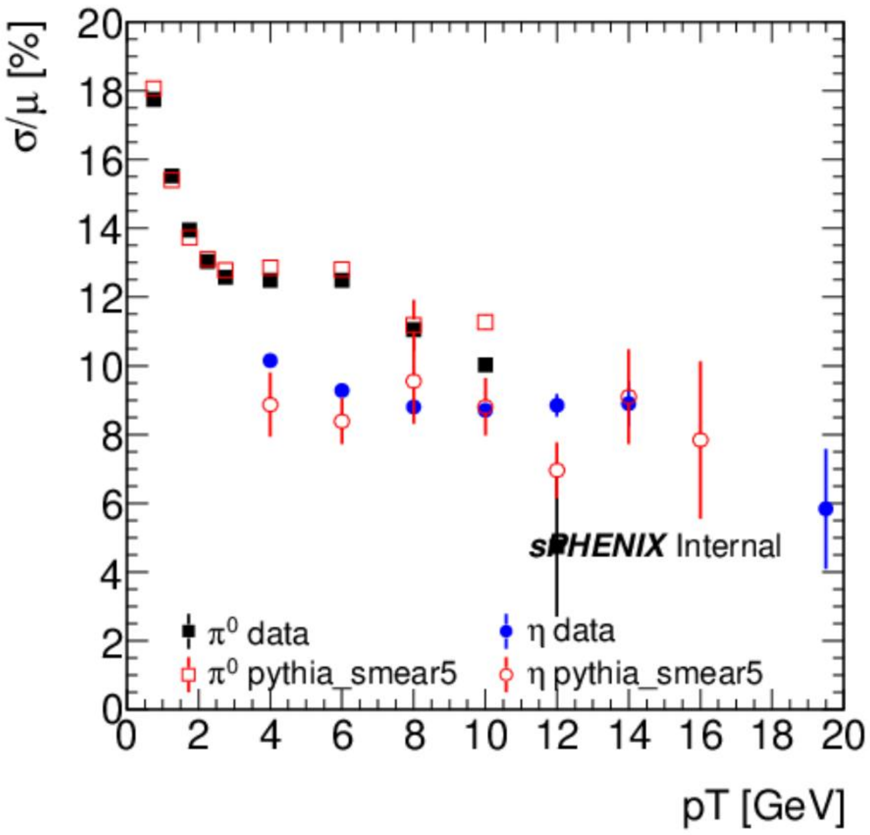




Post smearing (second attempt) data/MC

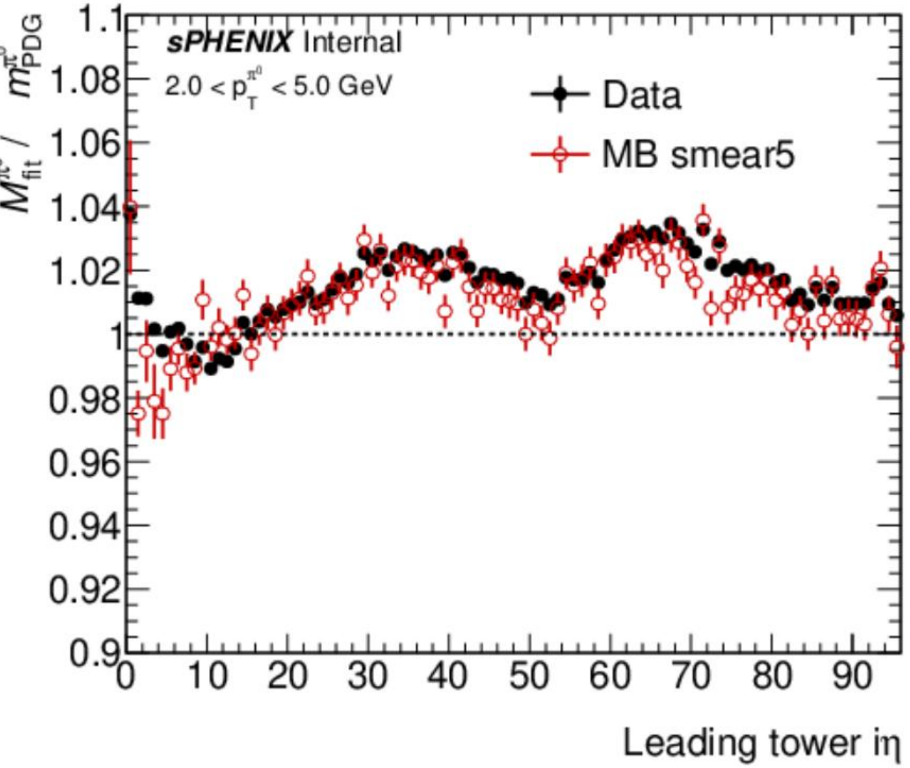
- Added an additional 8% of energy smearing
- Agreement in width is at the statistical limit of the MC
- Mass is shifted to agree
- Large discrepancy at high pT in the eta

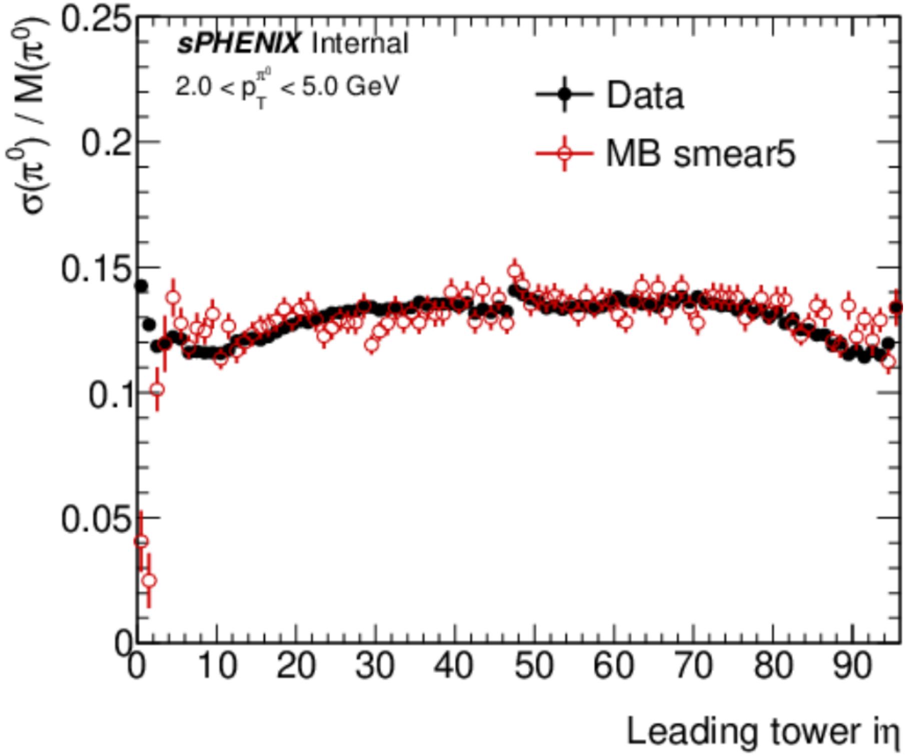


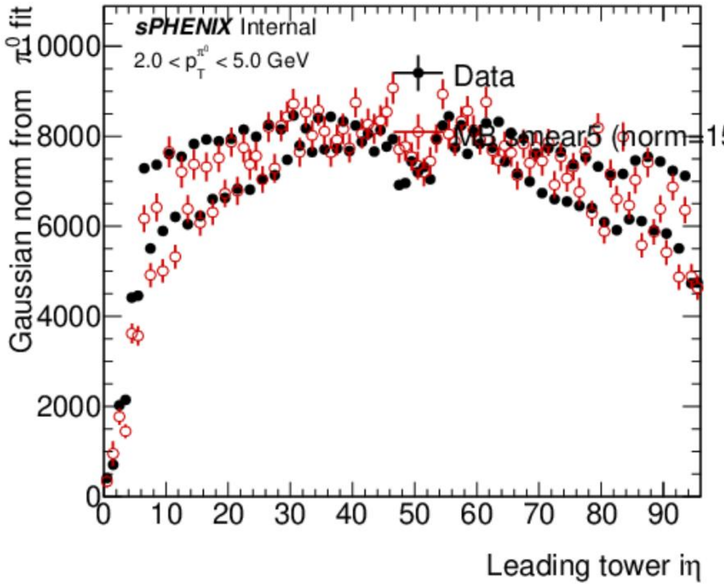


Calibration

➢ **Set scale eta-by-eta**

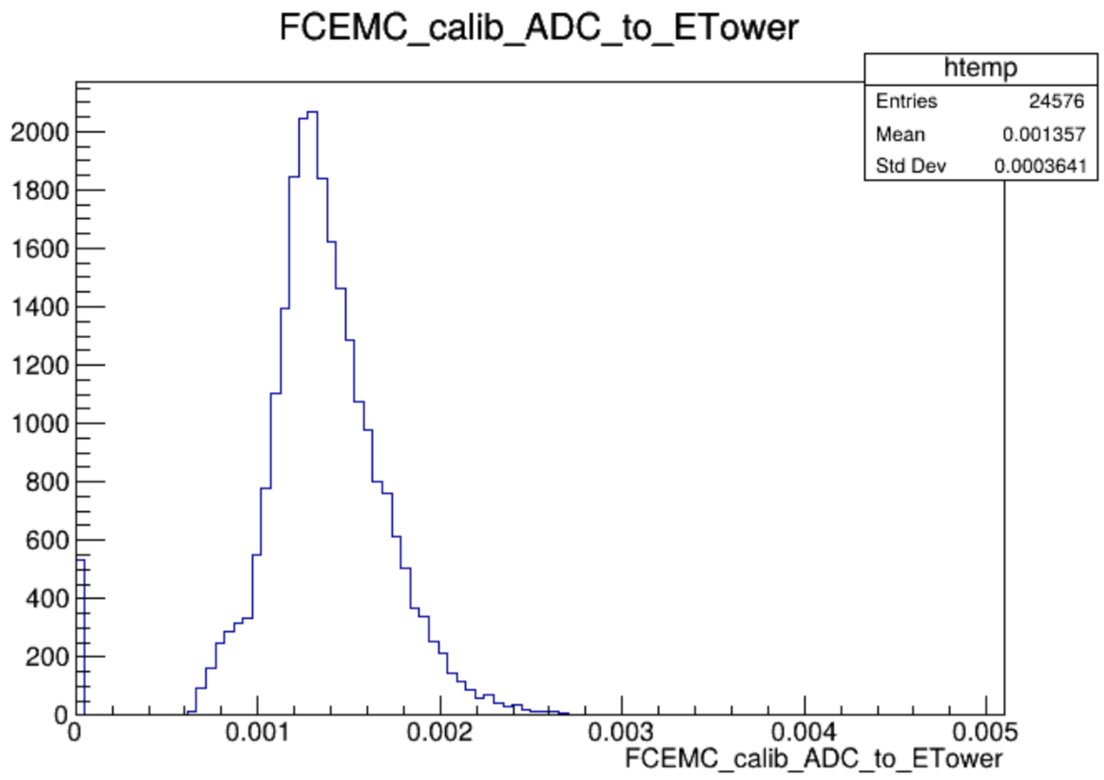


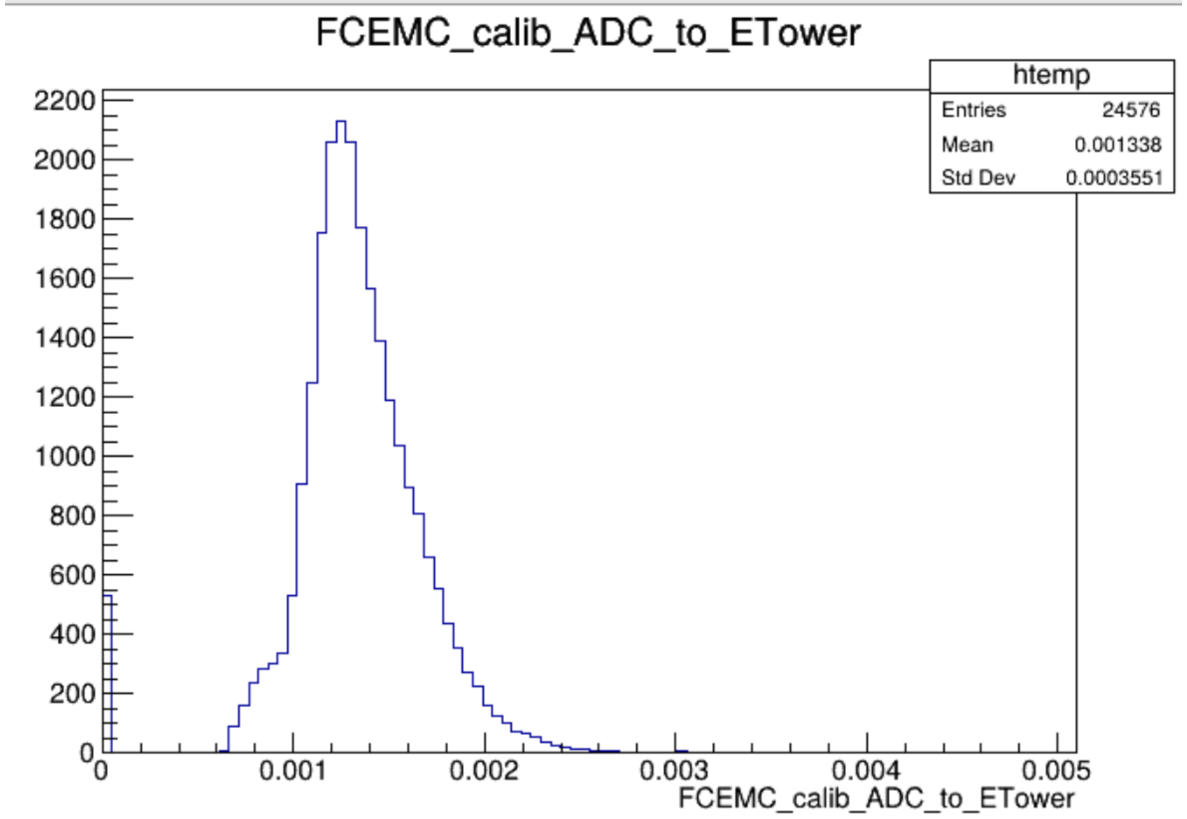




Calibration factors

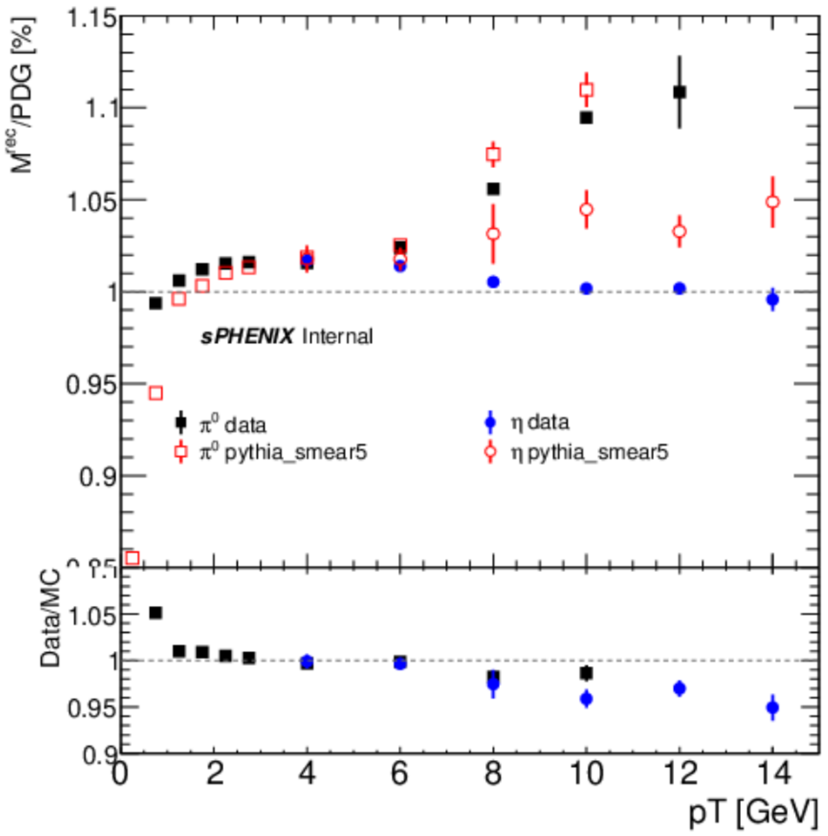
- **New** **old**

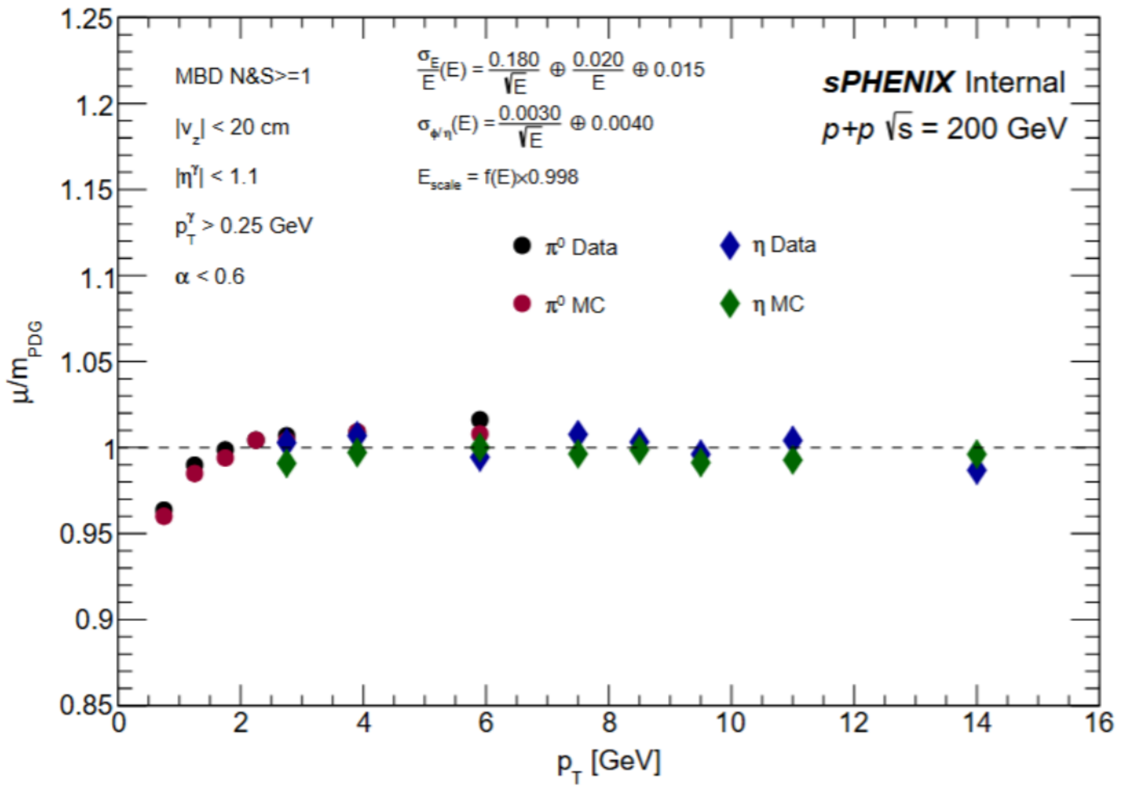




High pt eta meson

- ➢Right: Jaebeom's instance of the dame plot
- ➢Need more stats in MC



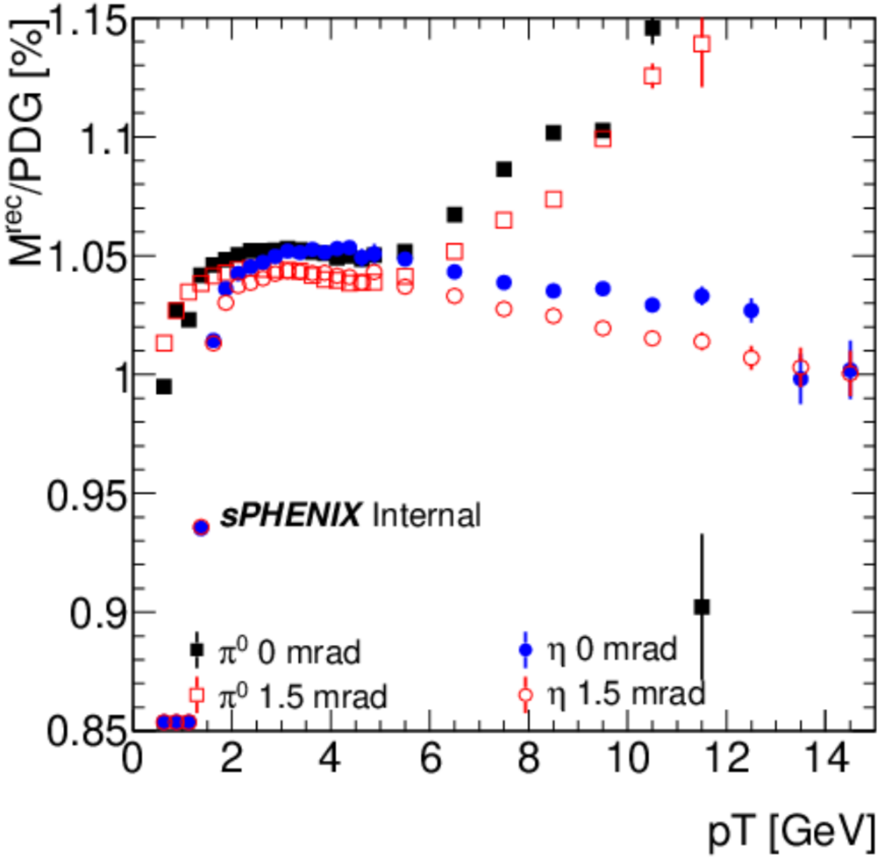


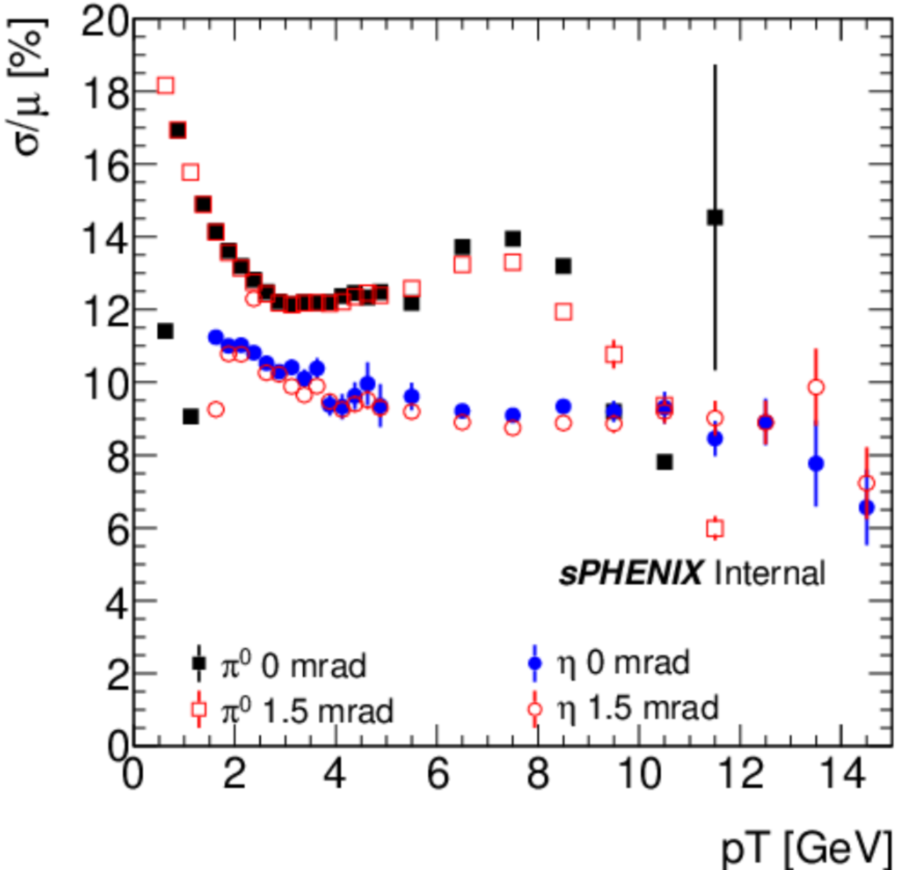
Conclusion

- Nominal pythia has muchbetter resolution than data
- It is unclear what is the major issue in data
- Extra smearing is helping describing the data but needs further tunning
- Need to also follow up on high pt eta as it is the only constraint on calibration at hight pT.

Omrad vs. 1.5 mrad data

- Omrad (1.5mrad) is estimated to have ~22% (7%) double interactions, which degrades vertex resolution
- This degradation is worse in 0mrad because the vertex width is !50 cm where it is ~15cm in 1.5mrad
- <u>For the rest of this presentation use 1.5 mrad for data/MC comparisons</u>





Pileup effects in MC

- Large effect seen in double interaction sim
- Could be effect of having the wrong vertex or could be poor performance at large truth z vertex which enters thebecause of a bad reco that is less than 20